LED Challenge 3 - Using a Button to Control Lights

Purpose: To program a button (input) to turn on a LED (output).

Materials: Computer, Arduino Uno Board, USB cable, Breadboard, LED, button, two 330 Ω Resistors (recommened), 2 Jumper Wires.

New Components:

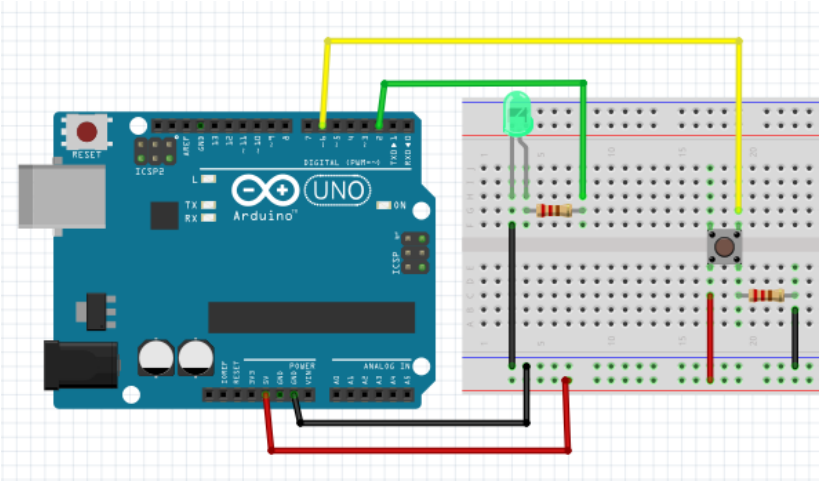
Push Button: A switch that can be used to signal a component to be turned on or off.



Resistors: Resistors resist the flow of electricity. You can use them to protect sensitive components like LEDs from too much current. The strength of a resistor (measured in ohms) is marked on the body of the resistor using small colored bands. Each color stands for a number, which you can look up using a resistor chart.

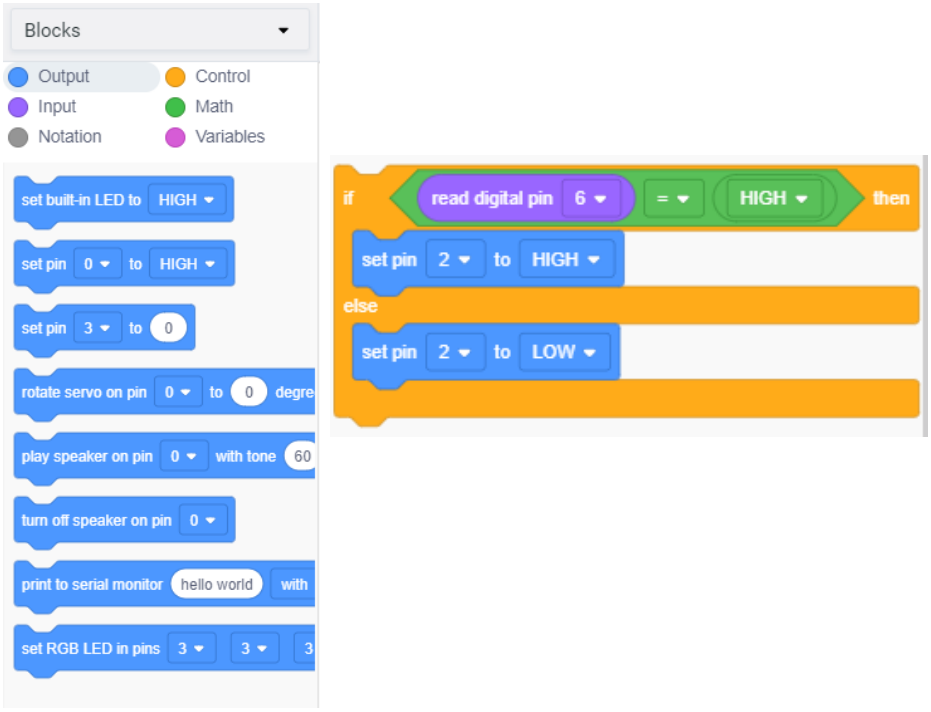
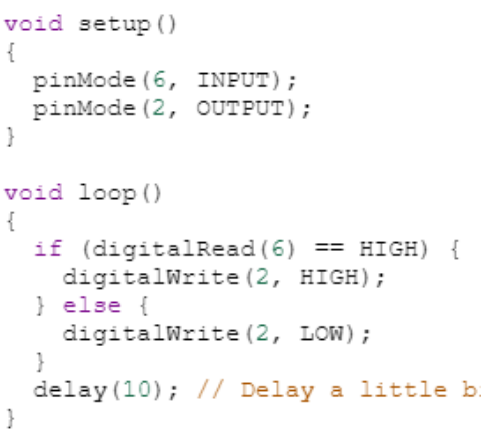
Procedure:

1. Wire the project using the following schematic.



2. You can create your code in the TinkerCad Editor then paste it to the **Arduino IDE**

3. Once in TinkerCad, use the following blocks and put them into the editor to get your button to control when the light is on and when the light is off. Watch the following video to help you with this.

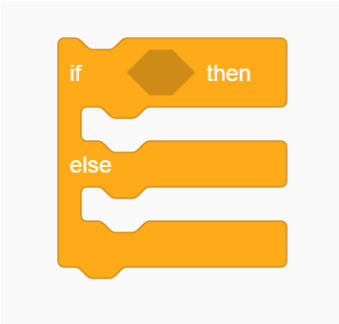
BLOCKS	ARDUINO
	


```
void setup()
{
  pinMode(6, INPUT);
  pinMode(2, OUTPUT);
}

void loop()
{
  if (digitalRead(6) == HIGH) {
    digitalWrite(2, HIGH);
  } else {
    digitalWrite(2, LOW);
  }
  delay(10); // Delay a little b
}
```

- In this code we are using the **"if...then"** conditional statement. The if statement checks for a condition and executes the proceeding statement or set of statements if the condition is 'true'. We can also use the **IF/ELSE** block if we want two possible decision branches.

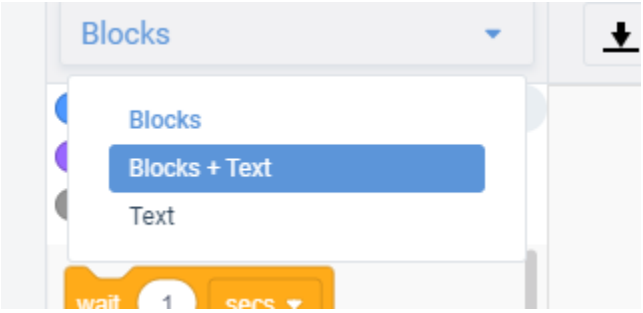




This means it will do the first set of blocks if the condition is true, and do the second set of blocks if it is not.

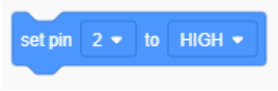
*****SAVE your code to your Robotics/Code Folder that you created*****

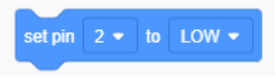
5. To see the text version of your code, pick “Blocks and Text” from the Code Menu drop-down:



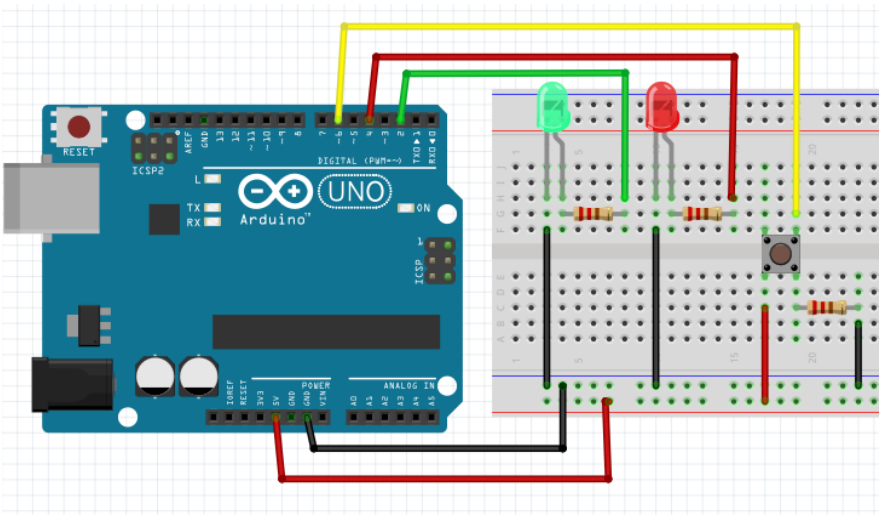
6. Copy the Arduino Code from TinkerCad into Arduino. Make sure to plug in the Arduino, and select the correct port (Tools → port)

6. Upload your program to the Arduino.

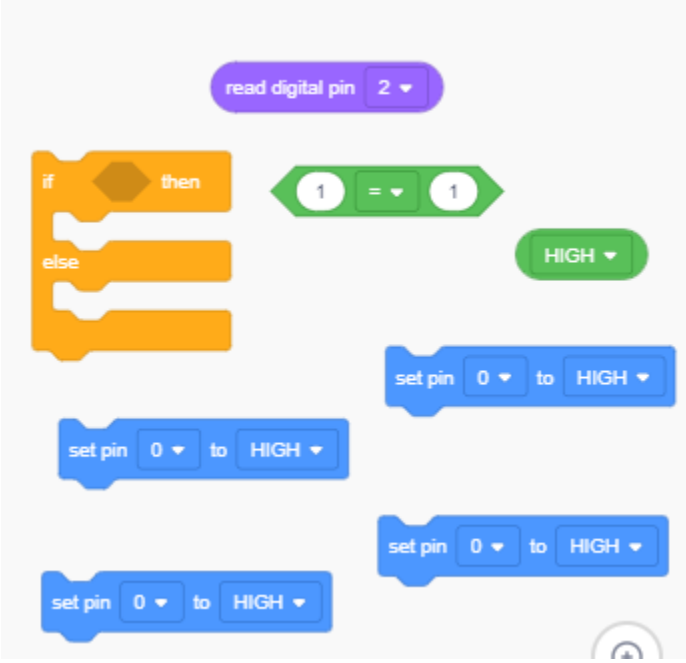
- Now that your code is uploaded observe the operation of the button. When the button is pressed, the light should turn on 

and when the button is released the light will turn off. 

7. Now you are going to add another light and a sequence to show how a light would work at a pedestrian crosswalk. When you press the button, the green light will go off for 5 seconds and the red light will go on to stop traffic so that it is safe to walk. The green light will go back on when it is no longer safe to walk.



8. Create the code to make your lights work using TinkerCad. You will need to use the following blocks.

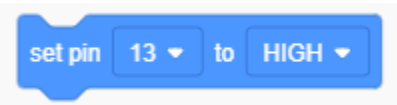


9. Once your program is loaded and your code and wiring is correct, your green light should be constantly on until you press the button. When the button is pressed, the green light will turn off and the red light will go on for 5 seconds

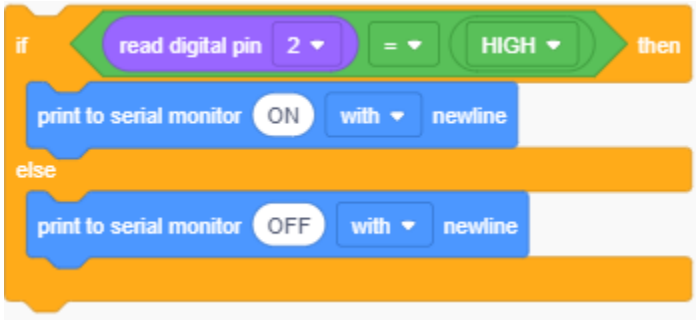
10. Observe your LEDs and complete your worksheet titled Assignment 1C: Using a Button to Control Lights

Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the leds, is the long positive arm connected by wires to the pins?
    c. Next inserting the LED directly into pin 13 and the adjacent ground. Does it light when the your code is on?



d. Next check the individual wires in the circuit. This can be done by connecting the LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
e. Check the button. This can be achieved by connecting it and using the serial monitor built into **the Arduino IDE.** Load this test program:



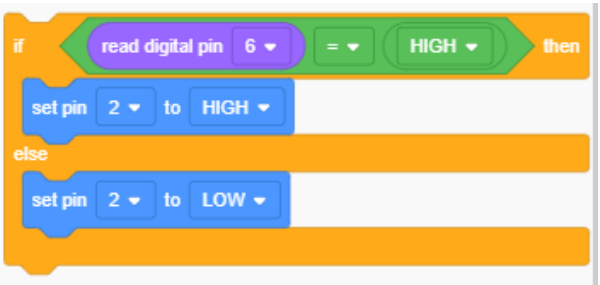
As you hold the button down the serial monitor should change to on from off.

Assignment 1C: Using a Button to Control Lights

1. In the first program, when you pressed the button what happened?

2. Why did you have to hold the button down in the first program to keep the light on? What block was added to the second program that allowed you to press and release the button to keep the red light on?

3. The following is a logic block that uses a conditional statement. Explain what the conditional statement is in this program.



Extension Challenge:

4. Modify your second program using the following blocks to make the red light blink on and off ten times at intervals of 0.1 seconds when the button is pressed.



5. The orange block is a loop. What is the purpose of a loop?

With funding from Canada

SIEC Saskatoon Industry Education Council

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